

Section D — AI-Augmented Learning

STEP 1 • BUILD WITH AI :-

Task 1 :- The AI-generated plan must specify the subnet, subnet mask, usable host range, and broadcast address

Ans.:-

Requirement	AI Plan
Subnet / Network	192.168.10.0/24
Subnet Mask	255.255.255.0
Usable Host Range	192.168.10.1 – 192.168.10.254
Broadcast Address	192.168.10.255

Task 2:- The plan must allocate addresses for all 5 systems and 1 printer named in the scenario.

Ans.:-

Network: 192.168.10.0/24

Subnet Mask: 255.255.255.0

Device Name	IP Address	Subnet Mask
System 1	192.168.10.10	255.255.255.0
System 2	192.168.10.11	255.255.255.0
System 3	192.168.10.12	255.255.255.0
System 4	192.168.10.13	255.255.255.0
System 5	192.168.10.14	255.255.255.0
Network Printer	192.168.10.15	255.255.255.0

Task 3:- Instruct the AI to also identify which addresses to reserve for future expansion.

Ans.:-

Future Expansion Reservation :-

The AI was instructed to reserve a range of IP addresses for future devices.

Reserved IP Range: 192.168.10.16 – 192.168.10.30

These 15 IP addresses are reserved for future expansion, such as additional workstations, printers, servers, or other network devices.

Task 4 :- Document the exact prompt(s) used.

Ans.:-

Create an IP addressing and subnetting plan for Nexus Innovations' startup office network with 5 computer systems and 1 network printer. Use a private Class C IPv4 subnet. Specify the subnet/network address, subnet mask, usable host range, and broadcast address. Assign a unique IP address to each of the 5 systems and 1 printer. Also identify and reserve a suitable range of unused IP addresses for future expansion. Present the complete plan in a clear table.

STEP 2 · TEST & DEBUG (WITHOUT AI) :-

Ans.:-

I manually tested the AI-generated IP addressing plan without using AI. I checked for duplicate or overlapping IP addresses, and all 5 systems and the printer had unique addresses. I verified that the network address is 192.168.10.0, the subnet mask is 255.255.255.0, the usable host range is 192.168.10.1–192.168.10.254, and the broadcast address is 192.168.10.255. The /24 subnet provides 254 usable hosts, which is sufficient for the six current devices. I also confirmed that 192.168.10.16–192.168.10.30 is reserved for future expansion. Therefore, no errors were found and no changes were required.